INCH-POUND

MIL-DTL-32630/3
3 July 2019

DETAIL SPECIFICATION SHEET

CONDUCTOR, ELECTRICAL, STRANDED, UNINSULATED CARBON-BASED CONDUCTIVE FIBER, -200 DEGREES CELSIUS to +200 DEGREES CELSIUS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-32630.

REQUIREMENTS:

TABLE I. Conductor properties.

Part or Identifying Number (PIN) (see 1/, 2/, 3/)	Conductor type (see 4/)	Coating type (see 5/)	Tow (no. of filaments)	Filament diameter (in)	Min. break load (lb.)	Min. elongation (%)	Max. DC resistance (Ω/ft.)	Max. linear density (lb./1000 ft)	Fiber CSA (in^2) (see 6/)
M32630/03-001	IV	N/A	1	0.006	0.98	12.7	93.10	0.007	2.82E-5
M32630/03-002	IV	N/A	1	0.009	3.83	3.0	11.81	0.011	5.02 E-5
M32630/03-003	IV	N/A	1	0.009	3.40	3.4	5.19	0.021	6.34 E-5

1/ M32630/03-001: 1 Ply 10Tex (mass in grams per 1000 meters) CNT Yarn
2/ M32630/03-002: 2 Ply 10Tex (mass in grams per 1000 meters) CNT Yarn
3/ M32630/03-003: 4 Ply 10Tex (mass in grams per 1000 meters) CNT Yarn
4/ Type IV - Carbon Nanotube Fiber (CNTF)
5/ N/A – Not Applicable. Conductor is not coated.
6/ Cross-sectional area (CSA) of the entire fiber.

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Conductors with the technical characteristics given in table I shall comply with the following requirements:

1. The minimum tensile strength shall be in accordance with table II.

TABLE II. Tensile strength.

Part or Identifying Number (PIN)	Minimum tensile strength (KSI)
M32630/03-001	27
M32630/03-002	76
M32630/03/003	127

2. Twists per inch: 8. When requested by the manufacturer, the qualifying activity can approve the use of different values.

3. Elongation: Minimum of 2.5 percent.

4. Balance ply: Required.

5. The temperature coefficient of resistance shall be in accordance with table III.

TABLE III. Temperature coefficient of resistance for conductors in table I.

Part or Identifying Number (PIN)	Temperature coefficient of resistance ($\Delta\Omega/\Omega$@20˚C)
M32630/03-001	-0.00048
M32630/03-002	0.00152
M32630/03-003	0.001694

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TABLE IV. Performance requirements for size 26 conductors.

Part or Identifying Number (PIN)	Conductor diameter (in.) (±.001)	Max. DC resistance conductor (Ω/ft)	Min. average break load (lbs.)	Elongation (%)	Max. linear density (lbs./1,000ft)	Min. fusing time (sec.) (at 2.5A)	Rated current (amperes)	Min. flex life (cycles)
M32630/03-004	0.0190	7.89	8.8	20	0.082	180	0.6	500
M32630/03-005	0.0170	1.55	18.6	4	0.099	22	0.84	
M32630/03-006	0.0180	1.52	21.3	5	0.090	22	0.88	

Conductors with the technical characteristics given in table IV shall comply with the following requirements:

1. The minimum tensile strength shall be in accordance with table V.

TABLE V. Tensile strength.

Part or Identifying Number (PIN)	Minimum tensile strength (KSI)
M32630/03-004	27
M32630/03-005	76
M32630/03/006	127

2. Twists per inch: 2. When requested by the manufacturer, the qualifying activity can approve the use of different values.

3. Elongation: Minimum of 2.5 percent.

4. Balanced ply: Required.

5. The temperature coefficient of resistance shall be in accordance with table VI.

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TABLE VI. Temperature coefficient of resistance conductors in table IV.

Part or Identifying Number (PIN)	Temperature coefficient of resistance ($\Delta\Omega/\Omega$@20˚C)
M32630/03-004	-0.00048
M32630/03-005	0.00152
M32630/03-006	0.001694

Notes:

1. Typical weight savings compared to copper: 90 percent.

2. Intended use. This conductor is intended for use in EMI shield and multi-strand conductor applications.

3. The conductor is manufactured in bulk lengths (hundreds to thousands of feet per lot) and supplied on bobbins in lengths for further processing (hundreds to thousands of feet per bobbin) in braiding, weaving or multi-stranding operations.

Custodians:
Army - CR
Navy - AS
Air Force - 85

Preparing activity:
Navy - AS

Project 6145-2018-053

Review activities:
Army - AR
Navy - SH
Air Force - 11, 20

NOTE: The activities above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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